VECTOR-BUNDLE DESCENT ACROSS A TWO-GENERATOR ALMOST-BLOWUP SQUARE AFTER FROBENIUS LOCALIZATION

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Editorial note. The square is exactly Exampletwo-generators, not an arbitrary proper square. The originalcolim_F category convention is included.

Editorial note. Public736 concerns pseudo-coherent derived triples across affine thickenings. This target concerns vector bundles across an almost blowup in characteristicp after Frobenius localization.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Characteristic-p Frobenius pullback, finite locally free modules, the explicitly defined two-generator almost-blowup square, Noetherian approximation, projective-line cohomology and formal functions. Function-levelFrobenius descent and thecolimF gluing lemma are retained.

Source reference. Original definition 0EU1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18536–18536. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 009F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 351–351. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0EVI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18588–18588. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03DT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4432–4432. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0EVG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18586–18586. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0EVJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18589–18589. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0B8W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14154–14154. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0E7H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17782–17782. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 02OC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3511–3511. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let p be a prime number. Consider an almost blowup square X, X', Z, E in characteristic p as in Example 0.2. Then the functor*

$$G : \operatorname{colim}_F \mathit{Vect}(X) \longrightarrow \operatorname{colim}_F \mathit{Vect}(Z) \times_{\operatorname{colim}_F \mathit{Vect}(E)} \operatorname{colim}_F \mathit{Vect}(X')$$

is an equivalence.

Example 0.2. Let A be a ring. Let $f_1, f_2 \in A$ be elements.

$$\begin{array}{ccc} E = \operatorname{Proj}(A/(f_1, f_2)[T_0, T_1]) & \longrightarrow & \operatorname{Proj}(A[T_0, T_1]/(f_2 T_0 - f_1 T_1) = X' \\ \downarrow & & \downarrow \\ Z = \operatorname{Spec}(A/(f_1, f_2)) & \longrightarrow & \operatorname{Spec}(A) = X \end{array}$$

is an almost blowup square.

1. Descent vector bundles in positive characteristic

A reference for this section is [1].

For a scheme S let us denote $\mathit{Vect}(S)$ the category of finite locally free $\mathcal{O}_S$-modules. Let p be a prime number. Let S be a quasi-compact and quasi-separated scheme over $\mathbf{F}_p$. In this section we will work with the category

$$\operatorname{colim}_F \mathit{Vect}(S) = \operatorname{colim} \Big(\mathit{Vect}(S) \xrightarrow{F^*} \mathit{Vect}(S) \xrightarrow{F^*} \mathit{Vect}(S) \xrightarrow{F^*} \ldots \Big)$$

where $F : S \to S$ is the absolute Frobenius morphism. In down to earth terms an object of this category is a pair $(\mathcal{E}, n)$ where $\mathcal{E}$ is a finite locally free $\mathcal{O}_S$-module and $n \geq 0$ is an integer. For morphisms we take

$$\operatorname{Hom}_{\operatorname{colim}_F \mathit{Vect}(S)}((\mathcal{E}, n), (\mathcal{G}, m)) = \operatorname{colim}_N \operatorname{Hom}_S(F^{N-n,*}\mathcal{E}, F^{N-m,*}\mathcal{G})$$

where $F : S \to S$ is the absolute Frobenius morphism of S. Thus the object $(\mathcal{E}, n)$ is isomorphic to the object $(F^*\mathcal{E}, n + 1)$.

Lemma 1.1. *Let p be a prime number. Let S be a scheme over $\mathbf{F}_p$. Let $(Sch/S)_h$ be a site as in Definition 0EU1. There is a unique sheaf $\mathcal{F}$ on $(Sch/S)_h$ such that*

$$\mathcal{F}(X) = \operatorname{colim}_F \Gamma(X, \mathcal{O}_X)$$

for any quasi-compact and quasi-separated object X of $(Sch/S)_h$.

Proof. Denote $\mathcal{F}$ the Zariski sheafification of the functor

$$X \longrightarrow \operatorname{colim}_F \Gamma(X, \mathcal{O}_X)$$

For quasi-compact and quasi-separated schemes X we have $\mathcal{F}(X) = \operatorname{colim}_F \Gamma(X, \mathcal{O}_X)$. by Sheaves, Lemma 009F and the fact that $\mathcal{O}$ is a sheaf for the Zariski topology. Thus it suffices to show that $\mathcal{F}$ is a h sheaf. To prove this we check conditions (1), (2), (3), and (4) of Lemma 0EVI. Condition (1) holds because we performed an (almost unnecessary) Zariski sheafification. Condition (2) holds because $\mathcal{O}$ is an fppf sheaf (Descent, Lemma 03DT) and if A is the equalizer of two maps $B \to C$ of $\mathbf{F}_p$-algebras, then $\operatorname{colim}_F A$ is the equalizer of the two maps $\operatorname{colim}_F B \to \operatorname{colim}_F C$.

We check condition (3). Let A, f, J be as in Example 0EVG. We have to show that

$$\operatorname{colim}_F A = \operatorname{colim}_F A/J \times_{\operatorname{colim}_F A/fA+J} \operatorname{colim}_F A/fA$$

This reduces to the following algebra question: suppose $a', a'' \in A$ are such that $F^n(a' - a'') \in fA + J$. Find $a \in A$ and $m \geq 0$ such that $a - F^m(a') \in J$ and $a - F^m(a'') \in fA$ and show that the pair (a, m) is uniquely determined up to a replacement of the form $(a, m) \mapsto (F(a), m+1)$. To do this just write $F^n(a'-a'') = fh+g$ with $h \in A$ and $g \in J$ and set $a = F^n(a') - g = F^n(a'') + fh$ and set $m = n$. To see uniqueness, suppose (a_1, m_1) is a second solution. By a replacement of the form given above we may assume $m = m_1$. Then we see that $a - a_1 \in J$ and $a - a_1 \in fA$. Since J is annihilated by a power of f we see that $a - a_1$ is a nilpotent element. Hence $F^k(a - a_1)$ is zero for some large k. Thus after doing more replacements we get $a = a_1$.

We check condition (4). Let X, X', Z, E be as in Example 0.2. By Lemma 0EVJ we see that

$$\mathcal{F}(X) = \operatorname{colim}_F \Gamma(X, \mathcal{O}_X) \longrightarrow \operatorname{colim}_F \Gamma(X', \mathcal{O}_{X'}) = \mathcal{F}(X')$$

is bijective. Since $E = \mathbf{P}^1_Z$ in this case we also see that $\mathcal{F}(Z) \to \mathcal{F}(E)$ is bijective. Thus the conclusion holds in this case as well. □

Lemma 1.2. *Let p be a prime number. Let S be a quasi-compact and quasi-separated scheme over $\mathbf{F}_p$. The category colim_F Vect(S) is equivalent to the category of finite locally free modules over the sheaf of rings $\operatorname{colim}_F \mathcal{O}_S$ on S.*

Proof. Omitted. □

Proof. Fully faithfulness. Suppose that $(\mathcal{E}, n)$ and $(\mathcal{F}, m)$ are objects of colim_F Vect(X). Let $(a, b) : G(\mathcal{E}, n) \to G(\mathcal{F}, m)$ be a morphism in the RHS. We may choose $N \gg 0$ and think of a as a map $a : F^{N-n,*}\mathcal{E}|_Z \to F^{N-m,*}\mathcal{F}|_Z$ and b as a map $b : F^{N-n,*}\mathcal{E}|_{X'} \to F^{N-m,*}\mathcal{F}|_{X'}$ agreeing over E. Choose a finite affine open covering $X = X_1 \cup \ldots \cup X_n$ such that $\mathcal{E}|_{X_i}$ and $\mathcal{F}|_{X_i}$ are finite free $\mathcal{O}_{X_i}$-modules. For each i the base change

$$\begin{array}{ccc} E_i & \longrightarrow & X_i' \\ \downarrow & & \downarrow \\ Z_i & \longrightarrow & X_i \end{array}$$

is another almost blow up square as in Example 0.2. For these squares we know that

$$\operatorname{colim}_F H^0(X_i, \mathcal{O}_{X_i}) = \operatorname{colim}_F H^0(Z_i, \mathcal{O}_{Z_i}) \times_{\operatorname{colim}_F H^0(E_i, \mathcal{O}_{E_i})} \operatorname{colim}_F H^0(X_i', \mathcal{O}_{X_i'})$$

by Lemma 1.1 (see proof of the lemma). Hence after increasing N we may assume the maps $a|_{Z_i}$ and $b|_{X_i'}$ come from maps $c_i : F^{N-n,*}\mathcal{E}|_{X_i} \to F^{N-m,*}\mathcal{F}|_{X_i}$. After possibly increasing N we may assume c_i and c_j agree on $X_i \cap X_j$. Thus these maps glue to give the desired morphism $(\mathcal{E}, n) \to (\mathcal{F}, m)$ in the LHS.

Essential surjectivity. Let $(\mathcal{F}, \mathcal{G}, \varphi)$ be a triple consisting of a finite locally free $\mathcal{O}_Z$-module $\mathcal{F}$, a finite locally free $\mathcal{O}_{X'}$-module $\mathcal{G}$, and an isomorphism $\varphi : \mathcal{F}|_E \to \mathcal{G}|_E$. We have to show that after replacing this triple by a Frobenius power pullback, it comes from a finite locally free $\mathcal{O}_X$-module.

Noetherian reduction; we urge the reader to skip this paragraph. Recall that $X = \operatorname{Spec}(A)$ and $Z = \operatorname{Spec}(A/(f_1, f_2))$, $X' = \operatorname{Proj}(A[T_0, T_1]/(f_2T_0 - f_1T_1))$, and $E = \mathbf{P}^1_Z$. By Limits, Lemma 0B8W we can find a finitely generated $\mathbf{F}_p$-subalgebra $A_0 \subset A$ containing f_1 and f_2 such that the triple $(\mathcal{F}, \mathcal{G}, \varphi)$ descends to $X_0 = \operatorname{Spec}(A_0)$ and $Z_0 = \operatorname{Spec}(A_0/(f_1, f_2))$, $X_0' = \operatorname{Proj}(A_0[T_0, T_1]/(f_2T_0 - f_1T_1))$, and $E_0 = \mathbf{P}^1_{Z_0}$. Thus we may assume our schemes are Noetherian.

Assume X is Noetherian. We may choose a finite affine open covering $X = X_1 \cup \ldots \cup X_n$ such that $\mathcal{F}|_{Z \cap X_i}$ is free. Since we can glue objects of $\operatorname{colim}_F \mathit{Vect}(X)$ in the Zariski topology (Lemma 1.2), and since we already know fully faithfulness over X_i and $X_i \cap X_j$ (see first paragraph of the proof), it suffices to prove the existence over each X_i. This reduces us to the case discussed in the next paragraph.

Assume X is Noetherian and $\mathcal{F} = \mathcal{O}_Z^{\oplus r}$. Using φ we get an isomorphism $\mathcal{O}_E^{\oplus r} \to \mathcal{G}|_E$. Let $I = (f_1, f_2) \subset A$. Let $\mathcal{I} \subset \mathcal{O}_{X'}$ be the ideal sheaf of E; it is globally generated by f_1 and f_2. For any n there is a surjection

$$(\mathcal{I}^n/\mathcal{I}^{n+1})^{\oplus r} = \mathcal{I}^n/\mathcal{I}^{n+1} \otimes_{\mathcal{O}_E} \mathcal{G}|_E \longrightarrow \mathcal{I}^n\mathcal{G}/\mathcal{I}^{n+1}\mathcal{G}$$

Hence the first cohomology group of this module is zero. Here we use that $E = \mathbf{P}^1_Z$ and hence its structure sheaf and in fact any globally generated quasi-coherent module has vanishing H^1. Compare with More on Morphisms, Lemma 0E7H. Then using the short exact sequences

$$0 \to \mathcal{I}^n\mathcal{G}/\mathcal{I}^{n+1}\mathcal{G} \to \mathcal{G}/\mathcal{I}^{n+1}\mathcal{G} \to \mathcal{G}/\mathcal{I}^n\mathcal{G} \to 0$$

and induction, we see that

$$\lim H^0(X', \mathcal{G}/\mathcal{I}^n\mathcal{G}) \to H^0(E, \mathcal{G}|_E) = H^0(E, \mathcal{O}_E^{\oplus r}) = A/I^{\oplus r}$$

is surjective. By the theorem on formal functions (Cohomology of Schemes, Theorem 02OC) this implies that

$$H^0(X', \mathcal{G}) \to H^0(E, \mathcal{G}|_E) = H^0(E, \mathcal{O}_E^{\oplus r}) = A/I^{\oplus r}$$

is surjective. Thus we can choose a map $\alpha : \mathcal{O}_{X'}^{\oplus r} \to \mathcal{G}$ which is compatible with the given trivialization of $\mathcal{G}|_E$. Thus α is an isomorphism over an open neighbourhood of E in X'. Thus every point of Z has an affine open neighbourhood where we can solve the problem. Since $X' \setminus E \to X \setminus Z$ is an isomorphism, the same holds for points of X not in Z. Thus another Zariski glueing argument finishes the proof. □

References

[1] Bhargav Bhatt and Peter Scholze. Projectivity of the Witt vector affine Grassmannian. *Invent. Math.*, 209(2):329–423, 2017.